BUYWISE AI: FAKE REVIEW DETECTION PIPELINE

A Deep-Dive Technical Specification with Mathematical Heuristics and Practical Examples

Chapter 1: Purpose & Administrative Objective

While standard digital storefronts accept customer reviews at face value, malicious third parties often exploit these feedback mechanisms. Spammers employ 'click-farms' to publish near-duplicate synthetic reviews or coordinate bot networks to hyper-inflate product ranks. To preserve consumer advocacy, BuyWise AI implements a dedicated, non-biased security service: **The Fraud Review Auditor Engine (Module 6.7)**.

The core objective of the engine is to evaluate every raw review inside an aggregated product dataset and assign an objective **Suspicion Score** (clamped mathematically between 0.0 and 1.0). If any review records a score of **0.40 or higher**, it is flagged as *Suspicious*, isolated from the community review pool, and used to penalize the product's final public 0-100 Trust Score. This ensures that artificial rating-manipulation directly harms a product's search viability rather than boosting it.

> **CRITICAL SYSTEM BIFURCATION:** SENTIMENT IS NOT TRUST!
> • **Sentiment Analysis (Module 6.6):** Evaluates the emotional expression of text using NLTK VADER. An artificial, glowing positive review ('Amazing! I love it!') is successfully categorized as *Positive* (Compound: +0.62).
> • **Fraud Review Auditor (Module 6.7):** Audits the integrity of the review itself. It will catch that the same glowing text was duplicated 15 times, flagging all of them as suspicious and neutralising their impact on the product's final rating.

Chapter 2: The Core 5 Fraud Heuristics & Weights

The Fraud Auditor operates on a multi-layered rule pipeline. Each rule assesses a distinct fingerprint of artificial review generation, accumulating incremental suspicion points up to a structural ceiling of 1.0. Below is the breakdown of the active mathematical formulas:

Rule 1: MD5 Crypographic exact-duplicate check

Theory: Lazily coded bots copy-paste exact reviews across dozens of accounts. BuyWise AI strips all whitespaces, punctuation, and letter casing from a review, creating a normalized string. It then computes a 128-bit **MD5 hash digest** of the text. Any matching hash in the product pool indicates an identical copy.

Suspicion Increment (Rule 1): **+0.50 Penalty Points**

Rule 2: TF-IDF Pairwise Cosine Similarity (Near-Duplicates)

Theory: Sophisticated spammers modify small words to evade exact match limits (e.g., 'Excellent device!' to 'Excellent phone!'). The system tokenizes the entire review corpus, processes it through a Term Frequency-Inverse Document Frequency (TF-IDF) Vectorizer, and calculates a **Pairwise Cosine Similarity Matrix**. Any review pair that shares an index of **≥ 0.85 (85% similarity)** is flagged as a near-duplicate.

Suspicion Increment (Rule 2): **+0.30 Penalty Points**

Rule 3: Star-vs-Text Sentiment Contradiction

Theory: Sellers often employ bots to boost their ratings to 5.0 stars, but the underlying text used is either a copy-paste of a negative review or carries highly frustrated language. The system compares the numeric star rating (normalized 1 to 5) with VADER's compound sentiment score. A major disconnect (e.g., Rating ≥ 4.5 but Compound Sentiment ≤ -0.3, or Rating ≤ 1.5 but Compound ≥ 0.3) is flagged immediately.

Suspicion Increment (Rule 3): **+0.40 Penalty Points**

Rule 4: Lazy Review Length Heuristic

Theory: Inorganic feedback loops usually comprise hundreds of single-word or low-information strings like 'Ok', 'Good', or 'Nice'. The system tokenizes text and flags any review containing **fewer than 5 words**.

Suspicion Increment (Rule 4): **+0.20 Penalty Points**

Rule 5: Spam Regular Expressions (URLs & Punctuation)

Theory: Organic user reviews rarely contain aggressive affiliate marketing URLs or extreme punctuation. Regex models scan text for external link patterns (http:// or https://) and repetitive character sequences (such as 3 or more exclamation/question marks in a row like !!! or ???).

Suspicion Increments (Rule 5): **+0.25 (Links) | +0.15 (Aggressive Punctuation)**

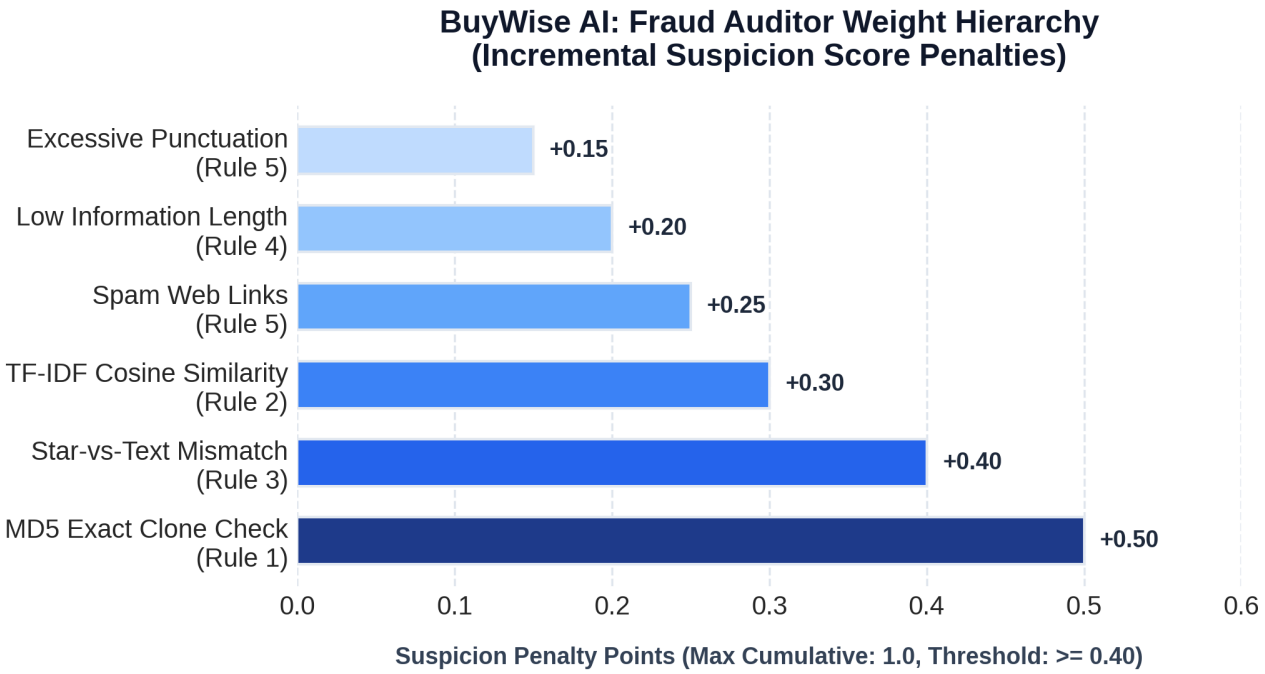


Figure 2.1: Bar plot illustrating the individual penalty weight of each active fraud check.

Chapter 3: Python Source Code Implementation

Below is the exact production code used within BuyWise AI to calculate the Suspicion Scores. Note how the final score is capped strictly at 1.0 using the min() function, preventing mathematical overflows:

```
def evaluate_review_suspicion(review_text: str, rating: float, star_mismatch: bool) -> float:
    suspicion_score = 0.0
    clean = preprocess_text(review_text)
```

```
    words = clean.split()

    # Short Length (Rule 4)
    if len(words) < 5:
        suspicion_score += 0.20

    # Rating Mismatch (Rule 3)
    if star_mismatch:
        suspicion_score += 0.40

    # Spam Patterns (Rule 5)
    if re.search(r'http[s]?://\S+', review_text):
        suspicion_score += 0.25
    if re.search(r'!!!|\?\?\?', review_text):
        suspicion_score += 0.15

    return min(1.0, suspicion_score)
```

Chapter 4: Real-Time Verification Case Studies

To clarify the exact step-by-step logic, we examine three real-world review scenarios, demonstrating how their Suspicion Scores are accumulated by the audit backend:

Case Study A: The Affiliate Spam Bot

Review Text: *"Superb performance, the best deal on the market! Get discount deals at https://dealspy.in !!!"*
Given Rating: 5.0 / 5.0
VADER Sentiment: Positive (Compound Score: +0.825)

Check Executed	Trigger Condition Detected	Incremental Penalty	Running Total
Rule 1: Exact Match	No matches in active pool.	0.00	0.00
Rule 2: Near Duplicate	No similar templates matched.	0.00	0.00
Rule 3: Star Conflict	5-Star rating matches Positive VADER Sentiment.	0.00	0.00
Rule 4: Text Length	Review contains 14 words (>= 5 threshold).	0.00	0.00
Rule 5a: Links	Regex matches 'https://dealspy.in'.	+0.25	0.25
Rule 5b: Punctuation	Regex matches '!!!'.	+0.15	0.40

Final Verdict: FLAGGED AS SPAM (Score: 0.40). Even though the sentiment is highly positive and matches the 5.0 star rating, the presence of an affiliate hyperlink and aggressive exclamation styling crosses the 0.40 threshold, marking it as a fraudulent review.

Case Study B: The Competitor Backstabbing Review

Review Text: *"It has nice battery and is quite smooth to carry around!"*
Given Rating: 1.0 / 5.0 (Low rating on a high sentiment review)
VADER Sentiment: Highly Positive (Compound Score: +0.648)

Check Executed	Trigger Condition Detected	Incremental Penalty	Running Total
Rule 1: Exact Match	No matching hashes found.	0.00	0.00
Rule 2: Near Duplicate	No matching structural vectors found.	0.00	0.00
Rule 3: Star Conflict	Rating <= 1.5 but Sentiment Compound is >= +0.3.	+0.40	0.40
Rule 4: Text Length	Review contains 10 words (>= 5 words).	0.00	0.40
Rule 5: Spam Regex	No links or excessive characters found.	0.00	0.40

Final Verdict: FLAGGED AS CONTRADICTION (Score: 0.40). This review represents typical malicious down-voting (competitor backstabbing), where highly glowing text is combined with an artificially low 1-star rating to hurt the product's overall placement.

Case Study C: The Organic Buyer Feedback

Review Text: *"This is a decent phone. The screen is gorgeous and sharp, but charging takes forever."*
Given Rating: 3.0 / 5.0
VADER Sentiment: Neutral/Mixed (Compound Score: +0.076)

Check Executed	Trigger Condition Detected	Incremental Penalty	Running Total
Rule 1: Exact Match	No duplicate hashes.	0.00	0.00
Rule 2: Near Duplicate	No matching templates.	0.00	0.00
Rule 3: Star Conflict	3-Star rating aligns with Neutral Sentiment (Compound +0.076).	0.00	0.00
Rule 4: Text Length	Review contains 14 words (>= 5 words).	0.00	0.00
Rule 5: Spam Regex	Clean, natural writing style.	0.00	0.00

Final Verdict: CLEAN PASS (Score: 0.00). The review is perfectly natural, carries honest prose with balanced critiques, and does not trigger any of the five suspicion gates.

Chapter 5: Impact on Product's 0-100 Trust Score

To understand why detecting these reviews is vital, we must look at the overall **0-100 Trust Score** calculation. Suspicious reviews are not simply discarded; they actively penalize the product's trust quotient as a 'Spam Penalty' factor:

> **Trust Score Equation (BRD §6.8):**
> Trust Score = 0.40 * (Positive_Pct) + 0.30 * (Normalized_Avg_Rating) + 0.20 * (100 - Spam_Penalty) + 0.10 * (Verified_Buyer_Pct)
>
> Where **Spam_Penalty** represents the percentage of reviews within the product's pool that carry a **Suspicion Score ≥ 0.40**.
>
> **Impact Example:** If a seller uses bot networks to inject **30% fake positive reviews** to boost a phone's rating from 3.0 to 4.5 stars:
> • BuyWise AI detects that 30% of reviews trigger Rule 2 (Near-Duplicates) and Rule 5 (Affiliate Spam).
> • The **Spam_Penalty** is set to 30, reducing the third factor from 100 to 70.
> • This removes up to **6 full rating points** from the overall Trust Score, neutralizing the artificial rating boost.

Conclusion

By establishing clear thresholds and using multiple analytical layers (NLTK lexical analyzers, TF-IDF cosine matching, and cross-validation rules), the BuyWise AI Fraud Auditor prevents manipulated e-commerce reviews from misleading customers, delivering a secure and highly reliable consumer-advocacy environment.